

Tutorial 5 – Advanced Function I

Pass by Value and/or return statement

1. You're simulating a withdrawal function of a bank account system in these two scenarios.
 - i) Create a **withdrawal** function that attempts to withdraw money. If there's enough balance, subtract and then display the new balance. If not, display an appropriate error message. Then write a **main** function which initialize the balance of the account to RM2000. Prompt user for the withdrawal amount required and invoke the **withdrawal** function.
 - ii) Create a **withdrawal** function that attempts to withdraw money. If there's enough balance, subtract and **return** the new balance. If not, print an appropriate error message. Then write a **main** function which initialize the balance of the account to RM2000. Prompt user for the withdrawal amount required and invoke the **withdrawal** function. For successful withdrawal, display the balance.
2. Develop a C++ function for a utility company that calculates monthly electricity bills for customers. Each customer pays a **base fee**, and additional units are charged based on the number of units consumed as per the table below:

<u>Units used</u>	<u>Rate per unit</u>
0-100	RM0.50
101-300	RM0.75
301 onwards	RM1.20

The function **calculateBill** will calculate and return the total bill based on the units used. The **main** function will prompt user for the units of electricity used, perform the **calculateBill** and display the bill charges accordingly.

3. A C++ program is designed to compute a customer's car rental charges. Every car is charged at RM25 per day with addition to a charge of RM 0.40 for every kilometre used. Finally, the final rental charges and the detailed breakdown of charges should be displayed on the screen.
 - (a) Draw a structure chart to represent the division of tasks above.
 - (b) Based on the structure chart, draw flowcharts to describe the solution of each task.
 - (c) Based on the flowchart given in (b), write a C++ program which integrates the necessary functions as described. Declare all the necessary constants, variables and parameters.

Pass by Reference

4. Create a C++ function **convertTime** that receives time in **hours** and returns both **minutes** and **seconds** using pass-by-reference. And then write a **main** function to prompt user for the time in hours and display the result of the time conversion in minutes and seconds.
5. Build a C++ function **updateItemPrice** for a shopping cart where it receives an item price and update it with a discount of 10% and then a 6% tax charges. The **main** function needs to prompt user for the item price, then update and display the price after discount and tax.

Pass by Address

6. In a banking system, two accounts accidentally got their balances swapped. Develop a C++ function **fixBalances** that takes two **addresses of double values** and swaps their values back to normal. Write a **main** function to display their balances before and after the swap.
Note: You may initialize the 2 account balances during the declaration.
7. Write a C++ function that takes the **length** and **width** of a rectangle, and returns the **area** and **perimeter** using pass by address. Then prepare a **main** function that prompts the user for the length and width of a rectangle, then find and display the area and perimeter.

Pass by Value, Reference, Address and/or return statement

8. Write a C++ function **processScore** that receives a student's **score (by address)** and returns the **grade (by reference)**. Add **5** bonus points to the score before determining the grade based on the following table:

Scores	Grade
85-100	A
70-84	B
60-69	C
50-59	D
0-49	F

Write a **main** function to get user to input the score, then find and display the final score after added 5) and grade of the student.

9. Build a C++ function that takes a **temperature in Celsius (by reference)** and returns the **Fahrenheit value (by address)**. Prepare a **main** function which the user inputs the temperature in Celsius, then convert and display it in Fahrenheit. The formula is as follows: -

$$\text{Fahrenheit} = (\text{Celsius} * 9/5) + 32$$

10. Write a C++ program that calculates the total rental cost for a car rental service. The program should include a function named **calculateRental** that accepts three parameters:

- The number of days the car is rented, which should be passed by address;
- A discount threshold, which is the number of days after which a discount applies, passed by value;
- A variable to store the discount amount applied, which should be passed by reference.

The function should use a **constant** daily rental rate of RM80 to compute the base rental cost. If the number of days rented exceeds the discount threshold, a 10% discount should be applied. The function should calculate and store the discount amount through the reference parameter and return the final total cost after applying the discount via the use of **return** statement.

In the **main()** function, prompt the user to enter the number of days the car is rented. Then, call the **calculateRental()** function with a chosen discount threshold (eg. 7 days), and display the discount amount and the final rental cost clearly to the user.